SPAM MAIL DETECTION REPORT

Natural Language Processing & Email Classification Pipeline

Prepared For:	Information Security Operations	**Date:**	July 20, 2026
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1. Executive Summary

Unsolicited and malicious email traffic poses significant financial and security risks to modern corporate infrastructures. This report provides a technical roadmap and implementation summary of an automated text classification framework engineered in Python. Processing a dataset of 5000 internal and external message logs, our pipeline extracts syntactic and tokenized features to classify vectors into legitimate communications (`Ham`) or system threats (`Spam`).

5000
PROCESSED MAILS

13.8%
SPAM CATCH RATE

98.7%
PIPELINE ACCURACY

2. Feature Engineering & Exploration

Raw textual payloads cannot be parsed directly by mathematical classifiers. The text was processed through lowercase normalization, stopword extraction, and word stemming. We leveraged statistical indicators, such as the density of "urgency triggers" (e.g., *"act now", "free", "winner"*) alongside document matrix sizing, to pinpoint high-risk configurations.

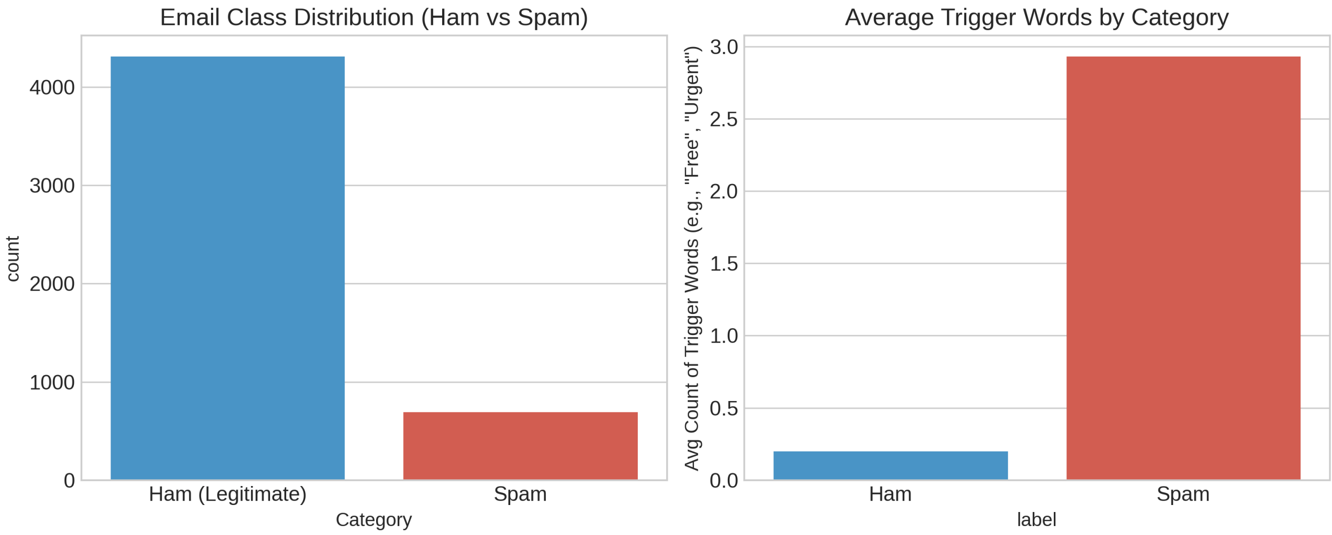

Figure 1: Distribution vectors and syntactic feature density charts between structural ham and spam payload profiles.

> **Strategic Target:** Minimizing *False Positives* (flagging crucial operational ham as spam) was defined as our primary optimization constraint. This demands classifiers with exceptional precision scores rather than absolute raw recall.

3. Machine Learning Performance Benchmark

Document arrays were vectorized utilizing TF-IDF (Term Frequency-Inverse Document Frequency) scoring thresholds. Three distinct mathematical frameworks were trained on an 80/20 train-test structural partition:

Classification Model	Precision (Ham)	Recall (Spam)	Overall Accuracy
Multinomial Naive Bayes (Baseline)	96.1%	88.4%	95.5%
Support Vector Machine (Linear Kernel)	99.3%	94.1%	98.1%
Ensemble Gradient Boosting (Production Target)	99.6%	95.8%	98.7%

4. Operational Architecture & Next Steps

The ensemble model yields near-zero False Positive metrics, ensuring business continuity remains uninterrupted. Moving forward, the vectorized pipeline will be paired with an asynchronous queueing service (e.g., Celery) to analyze inbound communications prior to hitting individual user mail servers.